Data Cleaning

DDD 3: Summer Field Practice Course

Ramesh Beniwal

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Cleaning of the field survey data from the urban slums: women, children and households. The code is shown as written; it is not run when the site is built.

Setup

```
rm(list = ls())
library(haven)
library(dplyr)
library(rio)
library(labelled)
library(ggplot2)
library(stargazer)
```

Clears the workspace and loads the packages.

Women's data: renaming variables

```
# 2. WOMEN_SLUMS

W_SLUMS <- W_SLUMS %>%
  rename(
    woman_index         = `_index`,
    date_interview      = Date_of_interview,
```

```
    team_id             = Team_ID,
    slum_id             = Slum_ID,
    household_id        = Household_ID,
    relation_hhd        = Relation_to_HHD_Head,
    joint_family        = Joint_family,
    fil_alive           = Father_in_law_alive,
    mil_alive           = Mother_in_law_alive,
    gender              = Gender,
    marital_status      = Marital_status,
    age                 = Age,
    age_at_marriage     = Age_at_marriage,
    education           = Education,
    currently_working   = Currently_Working,
    spouse_age          = Spouse_age,
    spouse_education    = Spouse_education,
    spouse_working      = Spouse_working,
    smartphone          = Smart_Phone,
    internet            = Internet,
    health_condition    = Health_condition,
    sleep_trouble       = Sleep_trouble,
    cognition           = Cognition,
    life_satisfaction   = Life_Satisfaction,
    depression          = Depression,
    d1                  = `group_wl2zg68/D1`,
    d2                  = `group_wl2zg68/D2`,
    d3                  = `group_wl2zg68/D3`,
    d4                  = `group_wl2zg68/D4`,
    d5                  = `group_wl2zg68/D5`,
    d6                  = `group_wl2zg68/D6`,
    d7                  = `group_wl2zg68/D7`,
    a1                  = `group_zx8ri59/A1`,
    a2                  = `group_zx8ri59/A2`,
    a3                  = `group_zx8ri59/A3`,
    a4                  = `group_zx8ri59/A4`,
    a5                  = `group_zx8ri59/A5`,
    mob_market          = `group_qv9zp21/Market`,
    mob_health          = `group_qv9zp21/Health_center`,
    mob_relative        = `group_qv9zp21/Relative_s_house`,
    mob_bank            = `group_qv9zp21/Bank`,
    mob_outside         = `group_qv9zp21/Outside_locality`,
    theft               = Theft,
    break_in            = Break_in,
    threat              = Threat,
```

```
    harassment          = Harassment,
    total_children      = Total_Children)
```

Gives short, clear names to the survey variables.

Women's data: cleaning

```
W_SLUMS <- W_SLUMS %>%
  mutate(
    age           = ifelse(age == 0, NA, age),
    education     = ifelse(education > 16, NA, education))
summary(W_SLUMS$age)
W_SLUMS |> count(education)
```

Age 0 and education above 16 are replaced with `NA`.

Women's data: recoding

```
W_SLUMS <- W_SLUMS %>%
  mutate(
    # widow or separated = 1, others = 0
    widow_sep = ifelse(marital_status %in% c(3, 4), 1, 0),
    # age groups
    age_group = case_when(
      age < 18               ~ 1,    # below 18
      age >= 18 & age < 25   ~ 2,    # 18-24
      age >= 25 & age < 35   ~ 3,    # 25-34
      age >= 35 & age < 45   ~ 4,    # 35-44
      age >= 45 & age < 55   ~ 5,    # 45-54
      age >= 55 & age < 65   ~ 6,    # 55-64
      age >= 65              ~ 7     # 65+
    ),

    # factors
    slum_id      = as.factor(slum_id),
    joint_family = as.integer(joint_family == 1),
    fil_alive    = as.integer(fil_alive == 1),
    mil_alive    = as.integer(mil_alive == 1))

W_SLUMS <- W_SLUMS %>%
  mutate(
    edu_cat = case_when(
      education == 0                   ~ 1,   # Illiterate
```

```
      education >= 1  & education <= 5  ~ 2,   # Primary
      education >= 6  & education <= 8  ~ 3,   # Middle
      education >= 9  & education <= 10 ~ 4,   # Secondary
      education >= 11 & education <= 12 ~ 5,   # Higher Secondary
      education >= 13 & education <= 16 ~ 6    # Graduate & Above
    ))
```

Widowed or separated indicator, age groups and education categories.

Dummy variables

```
# ── CREATE DUMMIES ──────────────────────────────────────────────────

W_SLUMS <- W_SLUMS %>%
  mutate(
    # education dummies
    edu_illiterate   = as.integer(edu_cat == 1),
    edu_primary      = as.integer(edu_cat == 2),
    edu_middle       = as.integer(edu_cat == 3),
    edu_secondary    = as.integer(edu_cat == 4),
    edu_higher_sec   = as.integer(edu_cat == 5),
    edu_graduate     = as.integer(edu_cat == 6),

    # age group dummies
    age_below_18     = as.integer(age_group == 1),
    age_18_24        = as.integer(age_group == 2),
    age_25_34        = as.integer(age_group == 3),
    age_35_44        = as.integer(age_group == 4),
    age_45_54        = as.integer(age_group == 5),
    age_55_64        = as.integer(age_group == 6),
    age_65plus           = as.integer(age_group == 7)
  )
```

One 0/1 variable for each education and age category.

Summary statistics

```
W_SLUMS %>%
  dplyr::select(
    # dependent variable
    currently_working,
```

```
    # independent variables — marital status
    widow_sep,

    # independent variables — family structure
    joint_family,

    # independent variables — age groups
    age_18_24,
    age_25_34,
    age_35_44,
    age_45_54,
    age_55_64,
    age_65plus,

    # control variables — education
    edu_illiterate,
    edu_primary,
    edu_middle,
    edu_secondary,
    edu_higher_sec,
    edu_graduate,

    # control variables — in-laws
    fil_alive,
    mil_alive
  ) %>%
  as.data.frame() %>%
  stargazer(
    type         = "text",
     title          = "Table 1: Summary Statistics — Determinants of Women's
Employment in Urban Slums",
    summary.stat = c("n", "mean", "sd", "min", "max"),
    digits       = 2,
    covariate.labels = c(
      # dependent
      "Currently Working",

      # independent
      "Widowed/Separated",
      "Joint Family",

      # age groups
      "Age 18-24",
```

```
      "Age 25-34",
      "Age 35-44 (ref)",
      "Age 45-54",
      "Age 55-64",
      "Age 65+",

      # education
      "Illiterate (ref)",
      "Primary (1-5 years)",
      "Middle (6-8 years)",
      "Secondary (9-10 years)",
      "Higher Secondary (11-12 years)",
      "Graduate & Above",

      # controls
      "Father-in-law Alive",
      "Mother-in-law Alive"
    ))
```

Summary table of the employment variables.

Importing and merging

```
W_SLUMS <- rio::import("../data/ddd3/SLUMS/W_SLUMS.dta")




# data merging
# 1. 1:1:1 merge (merge by a unique identifier)
#new_data<- merge(data1, data2, by = c("identifiers"))

# 2. m:1 merge
# new_data<- merge(data1, data2 by = c("identifiers"), all.x = TRUE)  : all.x =
TRUE means that all rows from data1 will be kept, and only matching rows from
data2 will be included.
# in this case we need to at place of data1  we need to keep many rows and at
place of data2 we need to keep one row.
```

Imports the women's data. The merge notes explain a one-to-one and a many-to-one merge.

Children's data

```
# 1 CHILD_SLUMS
library(tidyverse)
C_SLUMS <- C_SLUMS %>%
  rename(
    birth_order      = `children_history/C_Child_birth_order`,
    c_gender     = `children_history/C_Gender`,
    c_age        = `children_history/C_Age`,
    c_education  = `children_history/C_Education`,
    c_enrolled   = `children_history/C_currently_enrolled`,
    c_dropout    = `children_history/C_drop_out`,
    c_dropout_reason = `children_history/C_dropout_reason`,
    c_work       = `children_history/C_Work`,
    c_marital    = `children_history/C_Marital_Status`,
    parent_index     = `_parent_index`)
```

Renames the children's variables.

Household data

```
# 3. HH_SLUMS

HH_SLUMS <- HH_SLUMS %>%
  rename(
    hh_index          = `_index`,
    date_interview    = Date_of_interview,
    team_id           = Team_ID,
    slum_id           = Slum_ID,
    household_id      = Household_ID,
    hh_head_name      = Name_of_the_household_head,
    religion          = Religion,
    caste             = Caste,
    residence         = Residence,
    house_type        = House_type,
    n_rooms           = Number_of_rooms,
    rain_leak         = Rain_leak,
    hot_summer        = Hot_summer,
    ownership         = Ownership,
    electricity       = Electricity,
    water_supply      = Water_supply,
    toilet_facility   = Toilet_facility,
    cooking_fuel      = Cooking_fuel,
    ration_card       = Ration_card,
```

```
    ration_card_type  = Ration_card_type,
    land              = Land,
    tv                = `group_dd5nq69/A1`,
    fan               = `group_dd5nq69/A2`,
    cooler            = `group_dd5nq69/A3`,
    refrigerator      = `group_dd5nq69/A4`,
    washing_machine   = `group_dd5nq69/A5`,
    motorcycle        = `group_dd5nq69/A6`,
    bicycle           = `group_dd5nq69/A7`,
    auto_car          = `group_dd5nq69/A8`,
    smartphone        = `group_dd5nq69/A9`,
    computer          = `group_dd5nq69/A10`,
    sewing_machine    = `group_dd5nq69/A11`,
    livestock         = `group_dd5nq69/A12`,
    income_source     = Income_Source,
    food_exp          = Food_expenditure,
    non_food_exp      = Non_food_expenditure,
    income            = Income,
    migration         = Migration,
    loan              = Loan,
    loan_purpose      = Loan_purpose,
    loan_source       = Loan_Source,
    loan_amount       = Loan_Amount,
    awas_yojna        = `group_uz6ik40/Awas_Yojna`,
    ujjwala           = `group_uz6ik40/Ujjwala_Yojana`,
    pds               = `group_uz6ik40/PDS`,
    ayushman          = `group_uz6ik40/Ayushman_Bharat`,
    DAY               = `group_uz6ik40/Deendayal_Antyodaya_Yojana`,
    pm_svanidhi       = `group_uz6ik40/PM_Street_Vendor_Loan`,
    pension           = `group_uz6ik40/Pension`,
    swachh_bharat     = `group_uz6ik40/Swachh_Bharat_Mission`,
    ladli_behna       = `group_uz6ik40/Ladli_Behna_Yojana`,
    total_members     = Total_members,
    rain_leak_yes     = `Rain_leak/1`,
    rain_leak_no      = `Rain_leak/0`,
    hot_summer_yes    = `Hot_summer/1`,
    hot_summer_no     = `Hot_summer/0`,
    ownership_own     = `Ownership/1`,
    ownership_rent    = `Ownership/2`,
    ownership_informal = `Ownership/3`,
    ownership_other   = `Ownership/4`,
    cooking_lpg       = `Cooking_fuel/1`,
    cooking_elec      = `Cooking_fuel/2`,
```

```
    cooking_kerosene  = `Cooking_fuel/3`,
    cooking_firewood  = `Cooking_fuel/4`,
    cooking_other     = `Cooking_fuel/5`,
    loan_source_bank  = `Loan_Source/1`,
    loan_source_ml    = `Loan_Source/2`,
    loan_source_rel   = `Loan_Source/3`,
    loan_source_shg   = `Loan_Source/4`,
    loan_source_other = `Loan_Source/5`)
```

Renames the household variables.